MIL-P-13392B (MU)
21 February 1969
SUPERSEDING
MIL-P-13392A (MU)
3 April 1964 and
MIL-P-46955A (MU)
31 March 1964

MILITARY SPECIFICATION

PRIMER, PERCUSSION, M54
PARTS FOR AND LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the parts and the loading, assembling and packing for one type of percussion primer designated as Primer, Percussion, M54.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-D-46486 - Delay, Element, M2, Loading, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Lot Numbering of Ammunition

FSC: 1375

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MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

8840607 - Primer, Percussion, M54 Assembly
7548398 - Box, Set-up, Packing, Ammunition , Inner and Outer for Percussion Primers, M54 and T98
7548399 - Box, Packing, Ammunition for Percussion Primers M54 and T98

PUBLICATIONS

U. S. ARMY MUNITIONS COMMAND

ET - 8840607 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and parts.- The materials and parts shall be in accordance with applicable drawings (dwg.) and specifications.

3.2 Primer assembly.- The primer assembly shall comply with all requirements specified on Drawing (dwg.) 8840607, and with all requirements specified in applicable specifications.

3.3 Functioning

3.3.1 Sensitivity (see 6.4).- The primer assembly shall function by an applied energy of 15.68 inch ounces, maximum (max.) (see 6.5) and shall not function by an applied energy of 1.92 inch ounces, minimum (min.). The test shall be performed as specified in 4.4.1.1.

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3.3.2 Output.- The primer assembly, when assembled with the .050 second Delay Element (see MIL-D-46486) shall function the Delay Element, and the Delay Element shall burn not less than .020 seconds nor more than .070 seconds. The test shall be performed as specified in 4.4.1.2.

3.3.3 Duds.- The primer assembly shall function on the first attempt without evidence of a hangfire or squib (see 6.6), and the cup shall rupture sufficiently to expose the primer anvil when tested as specified in 4.4.1.3.

3.4 Workmanship.- All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. All packing components shall be dry.

3.5 First article inspection.- This specification makes provision for first article inspection. Requirements for the submission of first article samples shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

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4.1.1 Submission of product.- At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review, except that a certificate of analysis of the primer mixture shall be submitted with each lot submitted for acceptance.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.- The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of one hundred and twenty-five (125) each of every component part which has been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and he shall submit a record of this inspection with the sample, including certificates of conformance of materials. A

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sample containing known defects shall not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.- Components will be subjected by the Government to any or all of the applicable examinations and tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.- If any component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an item to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation.- The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of

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sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded primers shall contain:

a. Metal parts from not more than one lot interfix number from one supplier.

b. Primer mixture from not more than one batch or blend of batches which results in homogeneity of all composition used in the lot.

4.3.2 Examination.- The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cup (see dwg. 8840604 covering a detail of dwg. 8840607).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:			
1.	Cup cracked or split.....	Visual	01001
Major:	AQL 0.40 percent		
101.	Thickness through bottom.....	Gage	01002

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Minor:	AQL 1.50 percent		
201.	Outside diameter, max.....	Gage	01003
202.	Inside diameter.....	Gage	01004
203.	Length.....	Gage	01005
204.	Evidence of poor workmanship other than Critical above (see 3.4).....	Visual	01006

4.3.2.2 Anvil (see dwg. 8840605 covering a detail of dwg. 8840607).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Flash hole missing or obstructed.....	Visual	02001
102.	Location of flash hole.....	Gage	02002
Minor:	AQL 2.50 percent		
201.	Diameter.....	Gage	02003
202.	Thickness.....	Gage	02004
203.	Diameter of flash hole.....	Gage	02005
204.	Profile of convex end.....	Gage	02006
205.	Flat missing at concave end.....	Visual	02007
206.	Evidence of poor workmanship (see 3.4).....	Visual	02008

4.3.2.3 Primer assembly, prior to inserting disc (see dwg. 8840607).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Charge weight under minimum.....	Balance	03001
102.	Charge missing or incomplete....	Visual	03002

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Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4).....	Visual	03003

4.3.2.4 Primer assembly, prior to assembling anvil (see dwg. 8840607).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Loose composition on disc.....	Visual	04001
102.	Disc missing, damaged, improperly positioned or more than one.....	Visual	04002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4).....	Visual	04003

4.3.2.5 Primer assembly (see dwg. 8840607).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Cup cracked or split.....	Visual	05001
Major:	AQL 0.65 percent		
101.	Anvil loose, cocked, inverted, misformed or flash hole missing.	Visual	05002
102.	Cup dented or distorted.....	Visual	05003
103.	Loose composition on anvil or cup.....	Visual	05004
Minor:	AQL 2.50 percent		
201.	Diameter, max.....	Gage	05005
202.	Length.....	Gage	05006
203.	Stained or corroded components..	Visual	05007

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204.	Cup scratched.....	Visual	05008
205.	Evidence of poor workmanship other than Critical above (see 3.4).....	Visual	05009

4.3.2.6 Inner set-up box, prior to sealing (see dwg. 7548398).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Primer missing.....	Visual	06001
Minor:	AQL 1.50 percent		
201.	Tray damaged to extent that primers touch.....	Visual	06002
202.	Evidence of poor workmanship (see 3.4).....	Visual	06003
203.	Marking missing, incorrect or illegible.....	Visual	06004

4.3.2.7 Inner set-up box (see dwg. 7548398).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 1.50 percent		
201.	Box damaged to extent that contents are exposed.....	Visual	07001
202.	Marking missing, incorrect or illegible.....	Visual	07002
203.	Evidence of poor workmanship (see 3.4).....	Visual	07003

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4.3.2.8 Outer set-up box (see dwg. 7548398).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 1.50 percent		
201.	Sealing strip missing, mislocated or incompletely sealed.....	Visual	08001
202.	Box damaged to extent that contents are exposed or liable to become exposed.....	Visual	08002
203.	Label missing or unsealed.....	Visual	08003
204.	Marking incorrect or illegible...	Visual	08004

4.3.2.9 Outer set-up box in sealed barrier bag (see dwg. 7548398).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Bag cut or torn.....	Visual	09001
102.	Seal incomplete or mislocated....	Visual	09002
Minor:	AQL 0.65 percent		
201.	Marking missing, incorrect or illegible.....	Visual	09003

4.3.2.10 Packing box, prior to closing (see dwg. 7548399).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			

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Major:	AQL 0.40 percent		
101.	Outer set-up package missing....	Visual	10001
Minor:	AQL 0.65 percent		
201.	Filler missing.....	Visual	10002

4.3.2.11 Sealed packing box (see dwg. 7548399).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Box damaged to extent that contents are exposed.....	Visual	11001
102.	Strapping missing, broken or loose.....	Visual-Manual	11002
Minor:	AQL 1.50 percent		
201.	Strapping improperly assembled..	Visual	11003
202.	Marking missing, incorrect or illegible.....	Visual	11004
203.	Contents loose.....	Manual	11005
204.	Evidence of poor workmanship (see 3.4).....	Visual	11006

4.3.3 Testing.- Unless otherwise specified herein, the sampling plans for the following tests shall be in accordance with MIL-STD-105.

4.3.3.1 Functioning

4.3.3.1.1 Sensitivity (see 3.3.1) - Major defect - Code No. 05010.- A sample of 400 percussion primers shall be selected from each lot for this test. If the sum of the average critical height (H average) plus 5 times the standard deviation (S) multiplied by the weight of the ball exceeds the maximum requirement, or if the sum of the average critical height (H average) minus 2 times the standard deviation (S) multiplied by the weight of the ball is less than the minimum requirement, the lot shall be rejected (see 6.4). The test shall be performed as specified in 4.4.1.1.

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4.3.3.1.2 Output (see 3.3.2) - Major defect - Code No. 05011.- A sample of one hundred and twenty-five (125) primer assemblies shall be selected from each lot for this test. If two (2) or more primers fire but fail to initiate the delay element, or five (5) or more delay element assemblies fail to comply with the delay time requirements, the primer lot shall be rejected. If only one primer fires but fails to initiate the delay element, or more than two (2) but less than five (5) delay elements fail to comply with the delay time requirements, a second sample of one hundred and twenty-five (125) primer assemblies shall be selected and tested. If the combined number of defectives in both the first and second sample of delay element assemblies is two (2) or more primers that fire but result in delay element initiation failures, or seven (7) or more delay time failures, the primer lot shall be rejected. The test shall be performed as specified in 4.4.1.2.

4.3.3.1.3 Duds (see 3.3.3) - Major defect - Code No. 05012.- A sample of three hundred (300) primer assemblies shall be selected from each lot for this test. If two (2) or more assemblies fail to comply with the requirements, the lot shall be rejected. The test shall be performed as specified in 4.4.1.3.

4.3.3.2 Load test of cup (see dwg. 8840604) - Major defects

4.3.3.2.1 Light load - Code No. 01006.- A single sampling plan in accordance with Code Letter M and an AQL of 0.15 percent shall be used for this test. The test shall be performed as specified in 4.4.2.

4.3.3.2.2 Heavy load - Code No. 01007.- A single sampling plan in accordance with Code Letter M and an AQL of 0.15 percent shall be used for this test. The test shall be performed as specified in 4.4.2.

4.3.3.3 Heat seal test of barrier bag (see dwg. 7548398) - Major defect - Code No. 09004.- The method of testing, and acceptance and rejection shall be as specified in MIL-P-116.

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4.3.3.4 Composition of primer mixture (see dwg. 8840607) - Mjaor defect - Code No. 12001.- A sample of sufficient size to perform the composition determination shall be selected from each batch or blend of batches used in loading the primer lot. If the sample fails to comply with the requirements, the primer lot shall be rejected. The test shall be performed as specified in 4.4.3.

4.3.4 Inspection equipment.- Equipment Tabulation Number ET - 8840607 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.- Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures

4.4.1 Functioning.- The primers used in the following tests shall be conditioned at a temperature of seventy-five (75) degrees plus or minus five (5) degrees Fahrenheit (F.) for twenty-four (24) hours minimum prior to testing.

4.4.1.1 Sensitivity.- The sample of 400 primers shall be divided into eight groups of 50 primers each and tested in the following manner:

The first group (50) of primers shall be tested by dropping a steel ball 1.94 plus or minus .02 ounces on the firing pin of the test fixture from a height of 4 inches. (The height of drop shall be measured as the height from the top of the firing pin to the bottom of the ball.) The number of primers that fire, misfire, hangfire or squib (see 6.6) shall be recorded. This procedure shall be repeated at successively increasing heights until a height is reached at which all primers in the group being tested fire.

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The test fixture shall then be set at a height of 3 inches and the above procedure repeated at successively decreasing heights of one inch, using a new group of 50 primers at each height, until a height is reached at which all primers in the group being tested fail to fire. Acceptance or rejection of the lot shall be determined from calculations based on recorded data (see 6.4). All samples subjected to test shall be scrapped. Any remaining untested samples may be returned to the lot from which they were taken. The test equipment used shall be as specified in 4.3.4.

4.4.1.2 Output.- The primer assembly shall be assembled with the .05 second M2 Delay Element. The M2 Delay Element assembly shall be assembled in the firing fixture specified in 4.3.4 and fired by dropping a steel ball weighing 7.94 plus or minus .02 ounces on the firing pin from a height of ten (10) inches, max. The drop height is the distance between the bottom of the ball and the top of the firing pin. Observation shall be made for compliance with the requirements.

4.4.1.3 Duds.- The primer assembly shall be tested as specified in 4.4.1.1 using a drop height of eight (8) inches, max. Observation shall be made for compliance with the requirements.

4.4.2 Load test.- The cups shall be assembled in the test fixture specified in 4.3.4. The required light and heavy load shall be applied and the cups measured to determine compliance with the requirements. A different cup shall be used for each load application.

4.4.3 Determination of composition of primer mixture

4.4.3.1 Trinitrotoluene (TNT).- A known weight of approximately one gram (gm.) of the sample shall be transferred to a tared sintered glass or filtering crucible having a thick asbestos mat. Using a wooden spatula the sample shall be spread quickly in a thin layer over the bottom of the crucible, and weighed rapidly. The crucible shall be allowed to remain in a desiccator containing calcium chloride for at least 16 hours. To the dried sample in the crucible, 15 milliliters (ml.) of anhydrous chloroform shall be added. The solvent shall be allowed to stand in contact with the sample for 5 minutes. Then, with the end of a wooden rod, gently and carefully the lumps shall be crushed against the asbestos mat.

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The solvent shall be allowed to drain through the mat without suction. This extraction shall be repeated with two more 15 ml. portions of chloroform. The completeness of the extraction may be tested at this point by adding 2 drops of the chloroform filtrate to 2 drops of alcoholic potassium hydroxide solution (1 molar) on a white spot plate. A red coloration indicates incomplete extraction. If incomplete, the extraction shall be continued with 15 ml. portions of chloroform until extraction is complete, as shown by test. Air shall be drawn through the crucible until no odor of chloroform remains. The crucible and residue shall be dried in an oven, maintained at 100 plus or minus 5 degrees Centigrade (C.) for 15 minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated to percent trinitrotoluene on a moisture-free basis.

4.4.3.2 Antimony sulfide.- A 25 percent saturated ammonium acetate solution shall be prepared by diluting saturated ammonium acetate with 3 times its volume of water. The residue in the crucible from the trinitrotoluene determination shall be treated with 20 ml. of warm 25 percent saturated ammonium acetate solution. The solution shall be allowed to remain in contact with the residue for 10 minutes, breaking up any lumps with the flattened end of a glass rod. By means of suction the solvent shall be drawn off into a clean flask, and the extraction repeated with 2 more 20 ml. portions of the warm solution. Test for completeness of extraction of thiocyanate by collecting the last 1-2 ml. of the filtrate from the third treatment and to 1-2 drops add 2 drops of diluted nitric acid and 2 drops of slightly acidified ferric alum solution on a white spot plate. If extraction is not complete, as shown by a red coloration, treatment of the residue shall be continued with warm 10 ml. portions of the solvent. When extraction is complete, as shown by absence of red coloration when two drops of the filtrate are tested, the crucible and residue shall be washed thoroughly with water, collecting the washings with the ammonium acetate filtrate. The residue shall be washed with alcohol and ether, and aspirated until the odor of ether is no longer detectable. The crucible and contents shall be placed in an oven, maintained at 100 plus or minus 5 degrees C., for 15 minutes, then cooled in a desiccator and weighed. From this weight, the weight of the empty crucible shall be subtracted and the difference shall be calculated to percent antimony sulfide on a moisture-free basis.

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4.4.3.3 Lead thiocyanate.- The filtrate and washings from the antimony sulfide determination shall be transferred to a 400 ml. beaker. The volume should be between 150 and 200 ml. A 2 ml. portion of glacial acetic acid shall be added and the solution heated to boiling. While stirring vigorously, 8 ml. of 5 percent potassium dichromate solution shall be added slowly to the beaker. The solution shall be added slowly for 1 or 2 minutes longer, then digested on a steam bath for 1 hour, with frequent stirring. The precipitate should have a reddish-orange color. The precipitate shall be filtered through a tared filtering crucible, washed with hot water, and then crucible and precipitate shall be dried in an oven, maintained at 100 plus or minus 5 degrees C., cooled in a desiccator and weighed. The increase in weight shall be calculated to percentage of lead thiocyanate on a moisture-free basis as follows:

Lead Thiocyanate = 100.05 by weight of precipitate

4.4.3.4 Potassium chlorate plus gum.- The sum of the percentages of trinitrotoluene, antimony sulfide, and lead thiocyanate shall be subtracted from 100 and the remainder shall be reported as percentage of potassium chlorate plus gum. The percentage of potassium chlorate should be corrected by subtracting the gum percentage of 0.7 percent.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C.- The primers shall be preserved and packaged in accordance with dwg. 7548398.

5.2 Packing

5.2.1 Level C.- The outer set-up boxes shall be packed as specified in 7548399.

5.3 Marking.- Marking shall be in accordance with dwgs. 7548398 and 7548399.

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6. NOTES

6.1 Ordering data.- Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.- Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

6.2 Inspection code numbers.- The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.5).

6.3 Inspection equipment.- The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).- Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.- The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

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CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.- Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.- Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category A, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.5 Submission of contractor designs.- All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND15. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Formula and method of determining H average and s

6.4.1 Formula

H average = A plus B plus C
s = I times the square root of L minus M

where:
A = the sum of column 3.
B = the first drop height at which all elements did not fire.
C = one-half the drop height increment (see 6.7).
L = the sum of column 5.
M = the square of the sum of column 3.
I = the drop height increment.

6.4.2 Tabulations of data.- Divide a sheet of paper into 5 columns and tabulate the data as follows:

a. Title column 1 "Height of Drop" and list all drop heights in order, starting with the lowest height at which one or more primers fired and ending with the height at which one or more primers failed to fire.

DO NOT list the heights at which all elements fired or failed to fire.

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b. Title column 2 "Number of primers that failed to fire" and list the sum of the number of primers that failed to fire, squibbed or hangfired (see 6.6) at each intermediate height.

c. Title column 3 "Decimal Fractions of Elements Failing to Fire at Each Intermediate Height" and record this decimal fraction that failed to fire at each intermediate height. This number is obtained by dividing the value obtained under "Number of primers that failed to fire" by the number of elements tested at that height. Calculation should be carried out and recorded to the nearest one-hundredth (.10).

d. Title column 4 "Standard Deviation Factor" and list odd numerals in sequence starting with 1, for as many drop heights as are listed in column 1.

e. Title column 5 "Standard Deviation Factor Times Fraction of Failures" and, by multiplying each decimal fraction in column 3 by the number on the same line in column 4, enter the product on the corresponding line in column 5.

6.4.3 Procedure for solving equations.

a. To obtain "A" - Add the values in column 3 and substitute the sum for "A".

b. To obtain "B" - The first drop height at which all primers misfired.

c. To obtain "C" - Divide the drop height increment by 2 and substitute the dividend for "C".

d. To obtain "L" - Add the values in column 5 and substitute the sum for "L".

e. To obtain "M" - Square the sum of column 3 and substitute the square for "M".

f. To obtain "H average" - Add the values substituted for A, B and C.

g. To obtain "s" - Subtract the value substituted for "M" from the value substituted for "L" and then obtain the square root of the difference and then multiply the square root by the height of increment.

h. Using the calculated values for "H average" and "s", if H average plus 5"s" multiplied by the maximum weight of the ball is equal to or less than the maximum inch ounce energy value required, and "H average" minus 2"s" multiplied by the minimum weight of the ball is equal to or greater than the minimum energy value required, the lot should be considered acceptable for sensitivity.

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6.5 Inch ounces.- The drop height in inches multiplied by the weight of the ball in ounces.

6.6 Failure.- Defined as any one of the following primer malfunctions:

a. dud - a failure wherein initiation of the primer mixture does not occur.

b. squib - a failure wherein a burning of the primer mixture occurs without detonation producing a hissing or phizzing sound.

c. hangfire - a failure wherein an audible delay occurs between the instant of application of energy and the detonation report.

6.7 Height of increment.- The height of increment is defined as a constant interval in inches between successive drop heights which must be considered in calculating average height and the standard deviation in drop height.

6.8 Check test for possible deterioration.- The following test is designed to prevent loading of primers which have deteriorated into the ammunition component with which it is to be loaded. The tests should be performed on primers from each lot which have been stored for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the primers are loaded does not require that the check tests be performed, the contracting officer should insert provisions for the performance of the check tests in the contract. The contracting officer should arrange to have the contractor reimbursed for any expense incurred in the performance of the check test for possible deterioration.

6.8.1 Sensitivity.- The test shall be performed as specified in 4.4.1.1 using test equipment specified in 4.3.4. A sample of 600 primers shall be selected from each lot for this test. Seventy-five (75) primers shall be tested at each height. If H average plus 5.66s exceeds 8 inches or if H average minus 1.86s is less than 1 inch, the lot shall be rejected.

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6.8.2 Duds.- This test shall be performed as specified in 4.4.1.3. A sample of 315 primers shall be selected from each lot for this test. Failure of 3 or more primers to function shall reject the lot.

6.9 Submission of ammunition data cards.- In addition to the normal distribution of records, one copy of all ammunition data cards shall be forwarded to each of the following:

Commanding Officer
Picatinny Arsenal
Dover, New Jersey 07801
ATTN: (1) SMUPA-ND5
(2) SMUPA-DC.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1375-A-032

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 61 1426

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